VMware vSphere

Infrastructure Implementation Project

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System Administration Track

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1.Overview

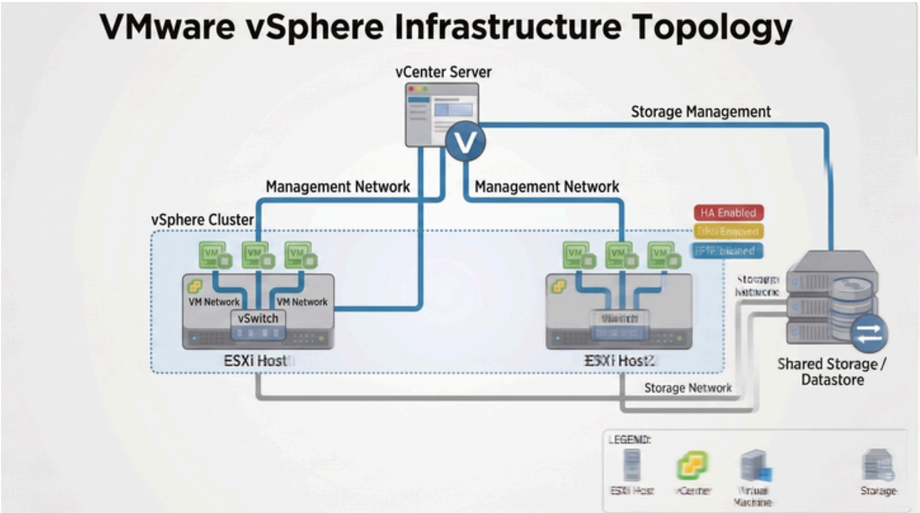
1.1 project objectives:-

The objective of this project is to design and implement a virtualized data center environment that demonstrates core enterprise virtualization concepts. This includes deploying multiple ESXi hosts, configuring centralized management through vCenter, and enabling key features such as High Availability (HA), Distributed Resource Scheduling (DRS), and Fault Tolerance (FT). The project aims to ensure efficient resource utilization, improve system reliability, and provide automatic recovery in case of host failure, while giving hands-on experience with real-world virtualization infrastructure and management.

1.2 project Architecture:-

This environment consists of two ESXi hosts running on separate physical machines,connected over a shared network via an access point
Center Server Appliance is deployed to centrally manage the hosts and enable clustering features such as HA and DRS. Shared storage is provided through an NFS server on a separate physical machine, allowing both hosts to access the same datastore. Network traffic is logically separated into management, vMotion, and storage networks.

Network Topology

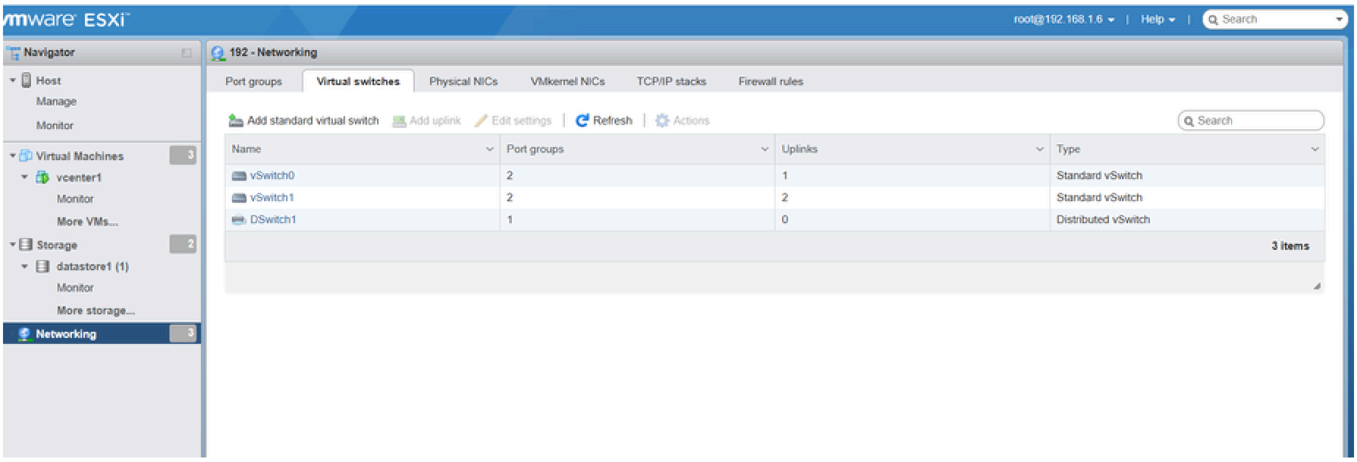


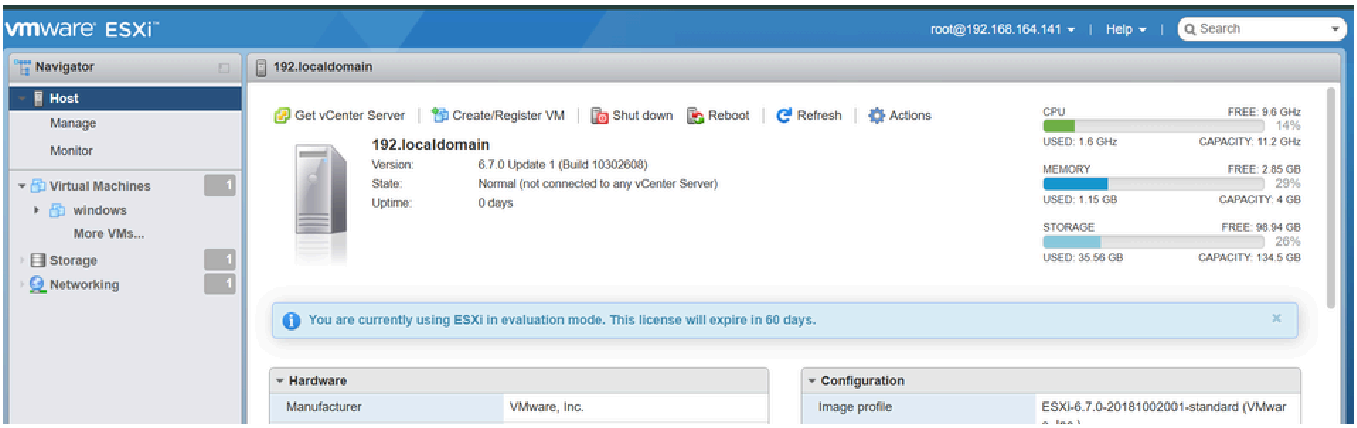
IP Address Assignment

DEVICE	IP ADDRESS
ESXI-01	192.168.1.6
ESXI-02	192.168.1.5
vCenter Server	192.168.1.11
NFS-Storage	192.168.1.14

2.Configuring Two ESXi Hosts as Virtual Machines

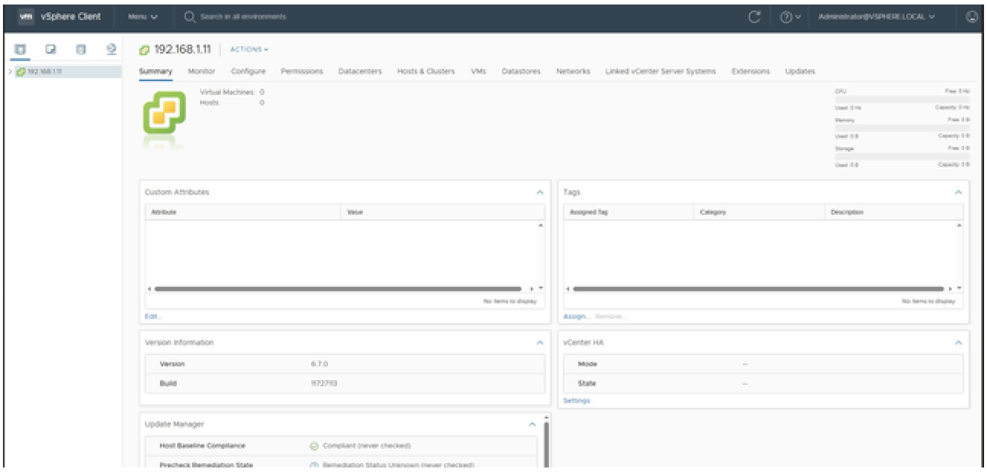
Two ESXi hosts were deployed as virtual machines using VMware Workstation. Each host was allocated adequate CPU, memory, and storage resources to support virtualization. In addition, each VM was configured with a network adapter set to Bridged mode, allowing it to obtain its own IP address from the access point it is connected to. The ESXi operating system was then installed and prepared for integration with vCenter Server.



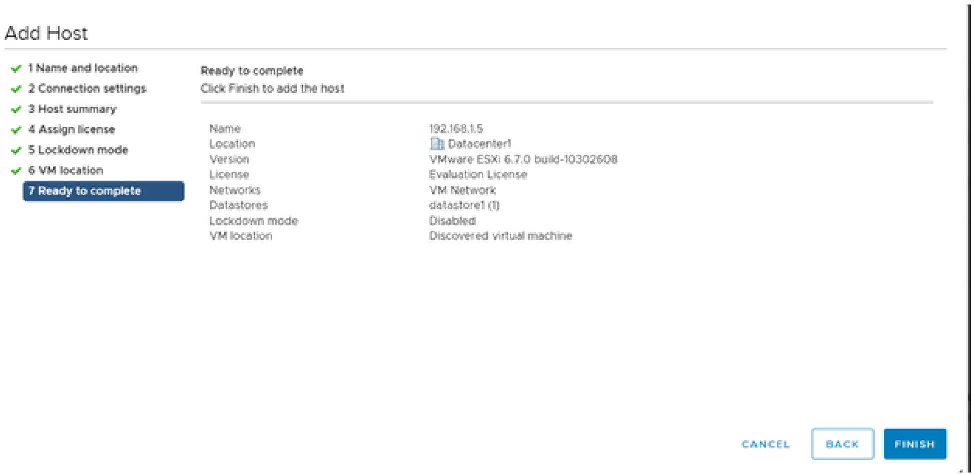


3. Setting Up vCenter Server and Integrating ESXi Hosts

3.1 Deploying the vCenter Server Appliance on One ESXi Host

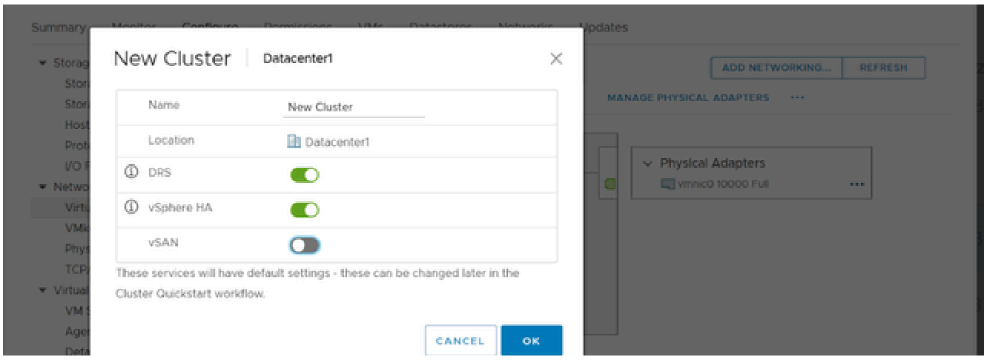


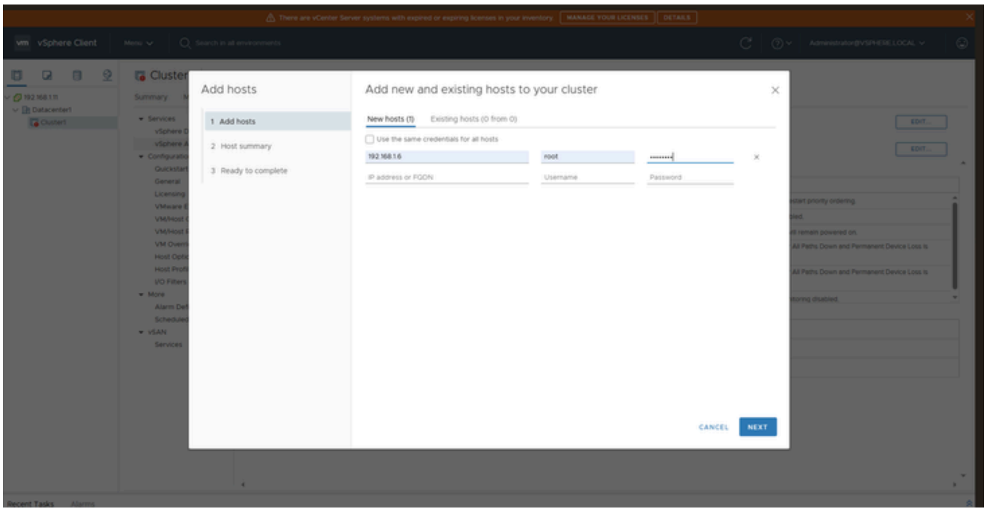
3.2 Adding the Second ESXi Host to vCenter

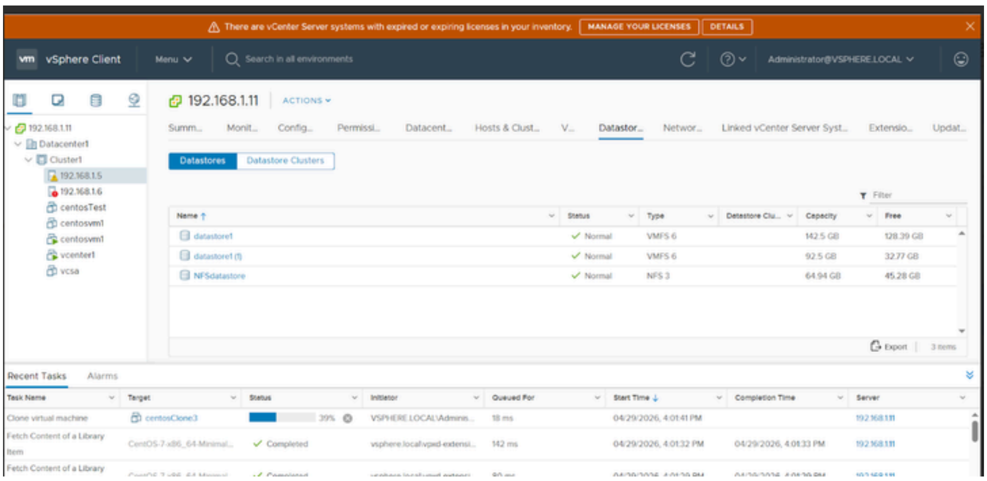


3.3 Creating a Cluster and Incorporating ESXi Hosts

After creating a Datacenter in vCenter Server, a cluster was configured to group the ESXi hosts into a single managed environment, enabling efficient resource sharing. Both ESXi hosts were then added to the cluster, and key features such as Distributed Resource Scheduling (DRS) and High Availability (HA) were enabled to ensure load balancing and automatic recovery in case of host failure



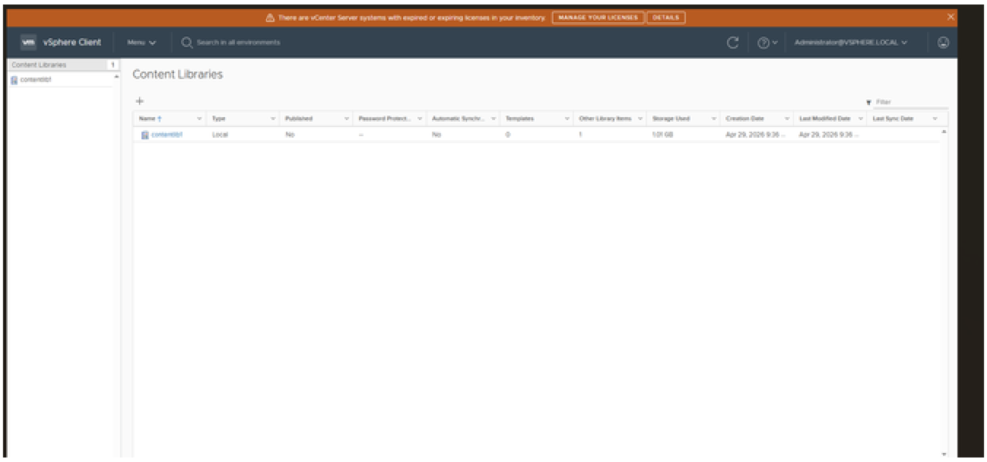




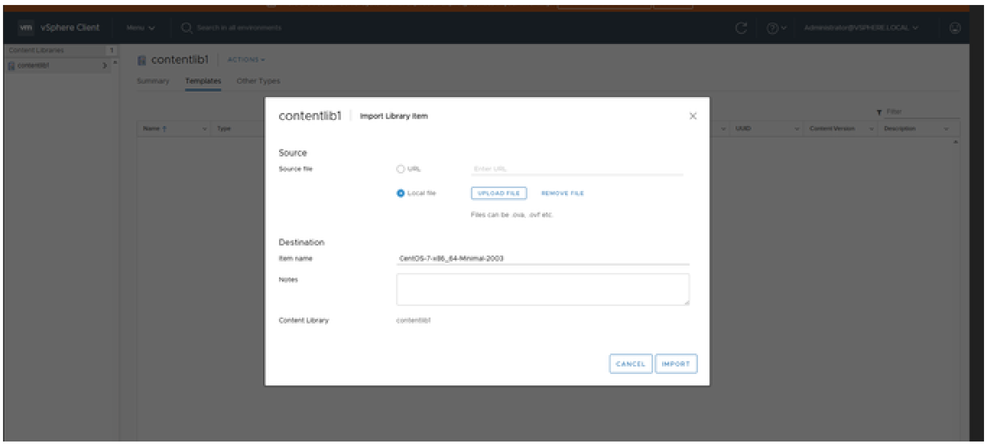
4.Content Library

4.1 setting up a content lib

Content Library was created in vCenter to centrally store and manage files such as ISO images and VM templates. This allows easy sharing of resources across ESXi hosts and simplifies virtual machine deployment and management.



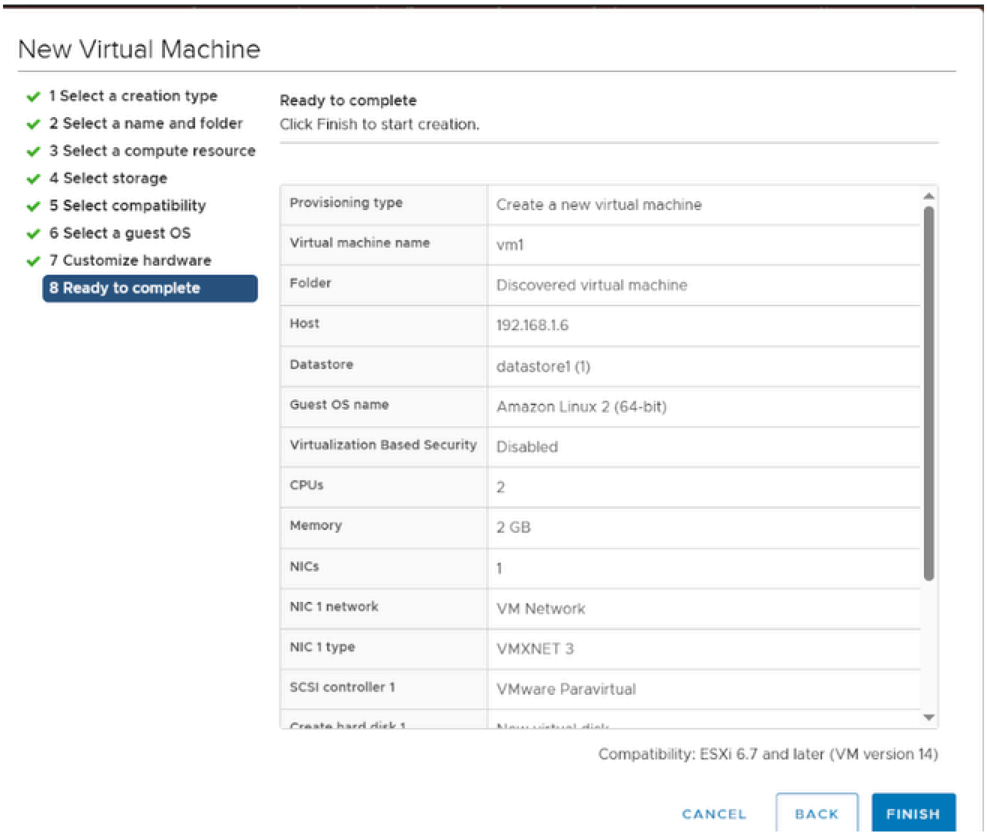
4.2 Uploading ISOs to the Content Library



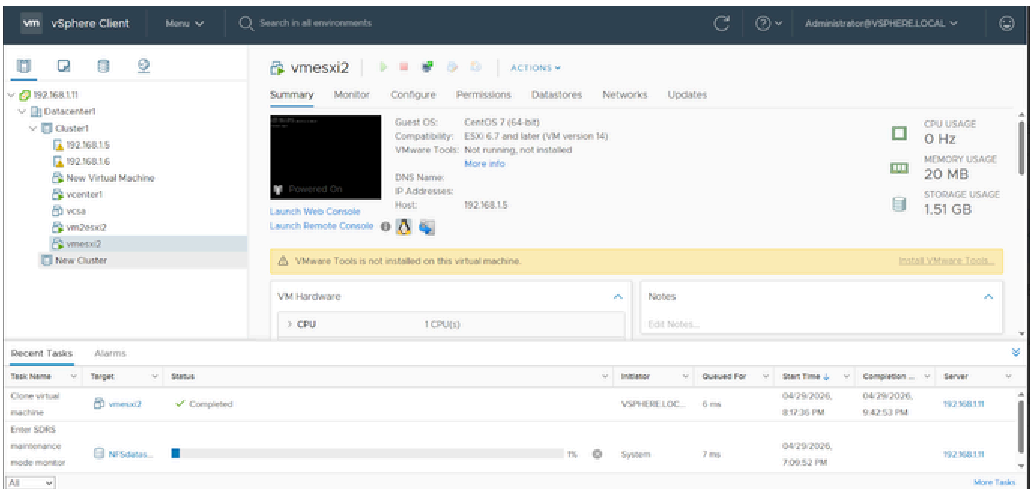
5. Virtual Machine Operations

5.1 Creating a Virtual Machine:

we created a linux virtual machine on esxi 1 using an iso we uploaded in the content lib

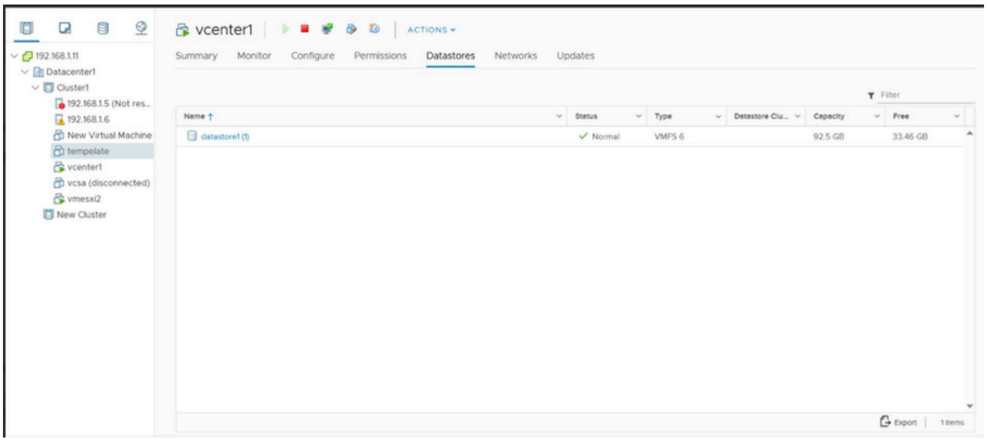


5.2 Cloning a Virtual Machine

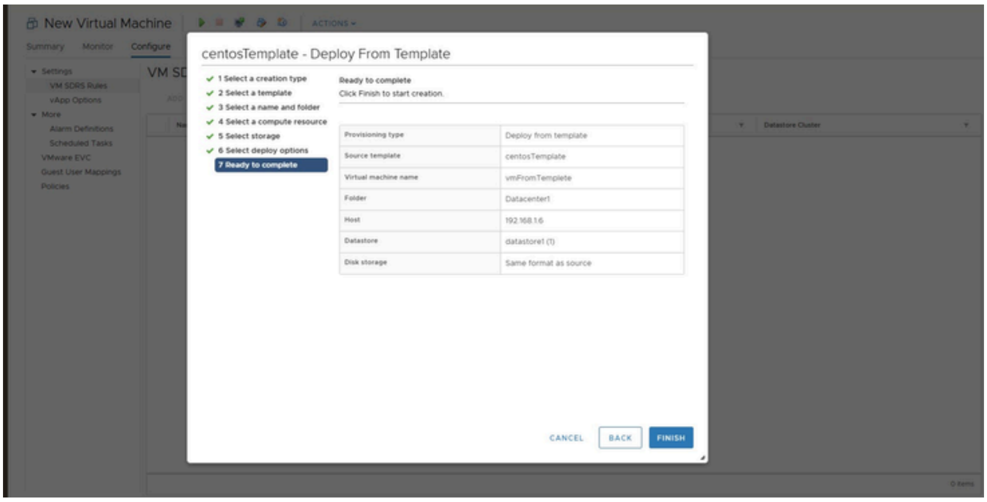


5.3 Generating a Template from a Virtual Machine:

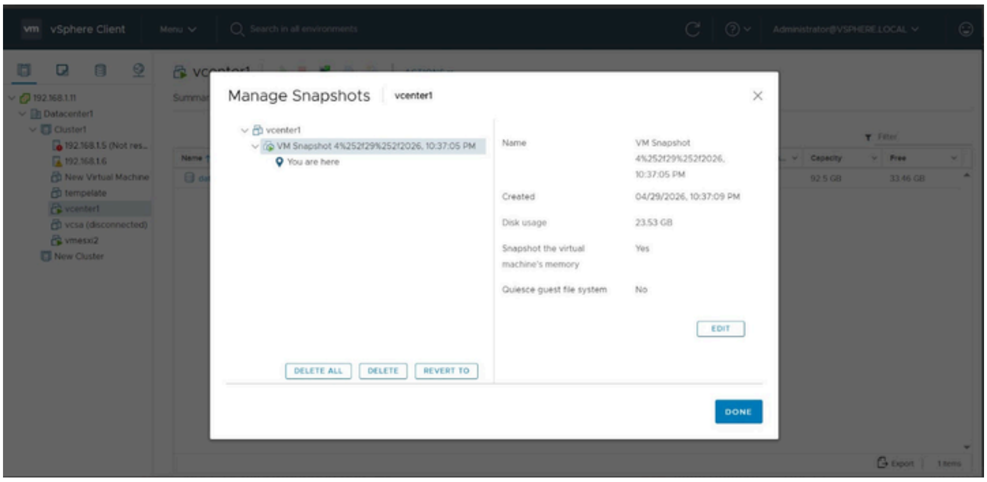
we created a template from the linux vm to be able to use it later



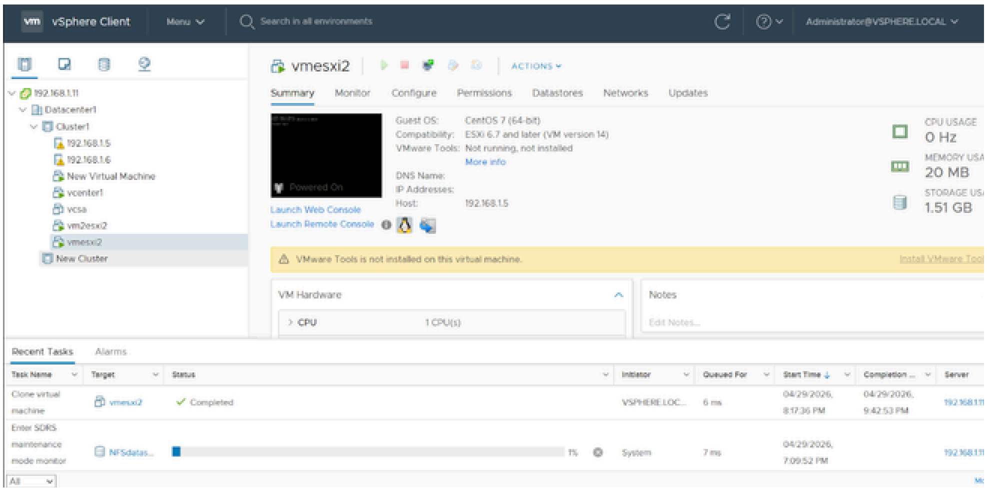
5.4 deploy vm from that template



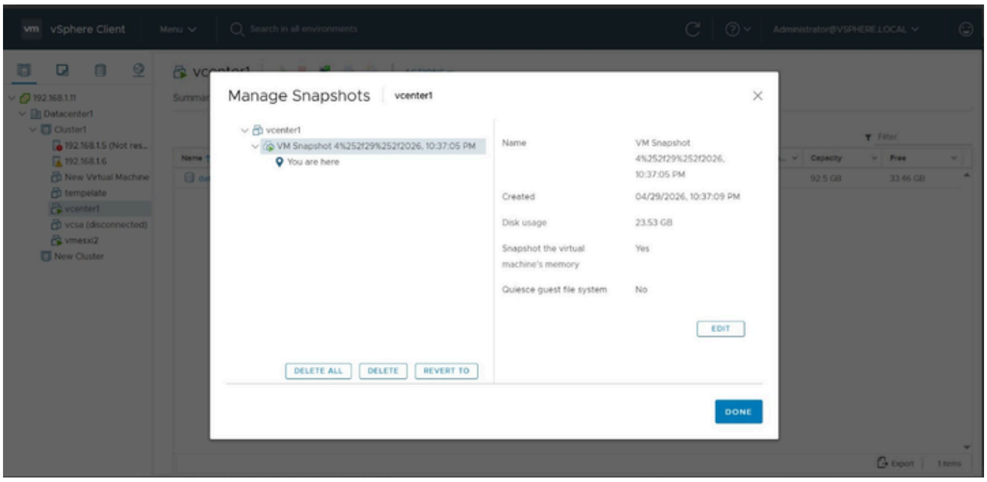
5.5 taking a snapshot from the vm



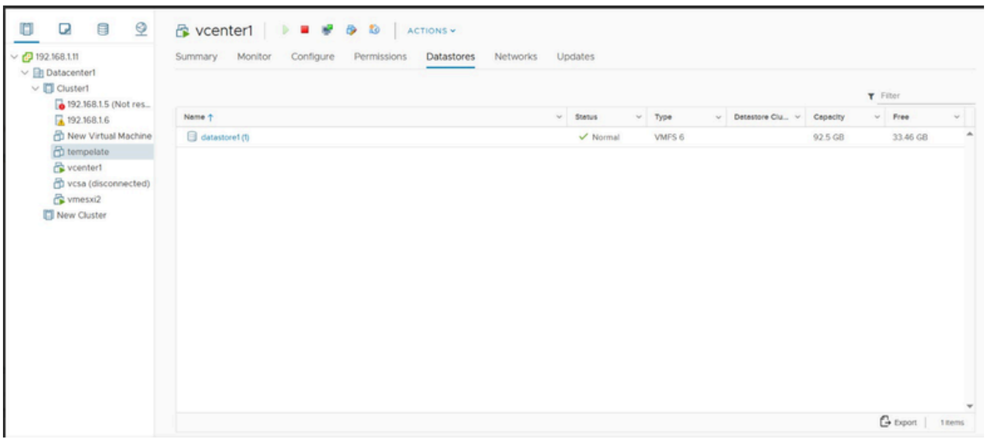
7.Perform operations like cloning, snapshotting, and template creation on you VM



Cloning a vm



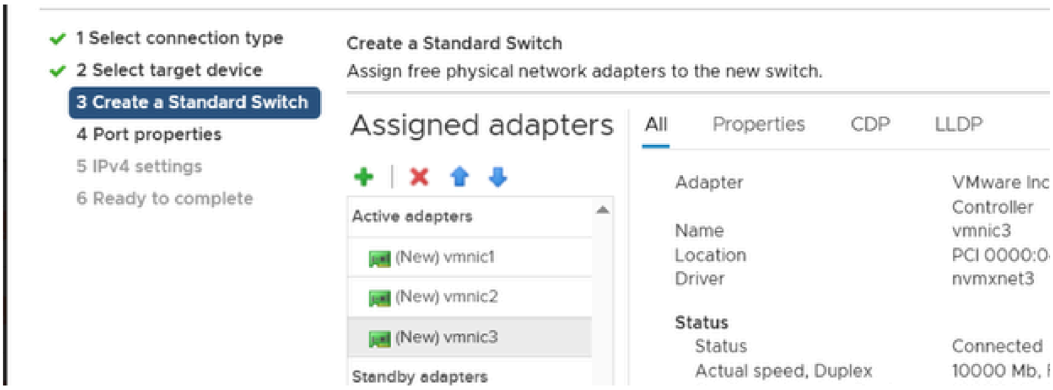
taking a snapshot from vm



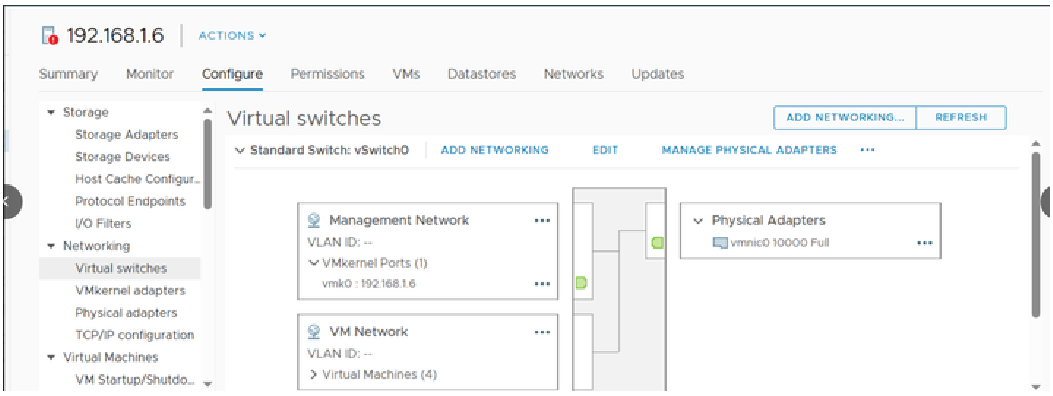
Create a Template from vm

6.Configuring Virtual Switches

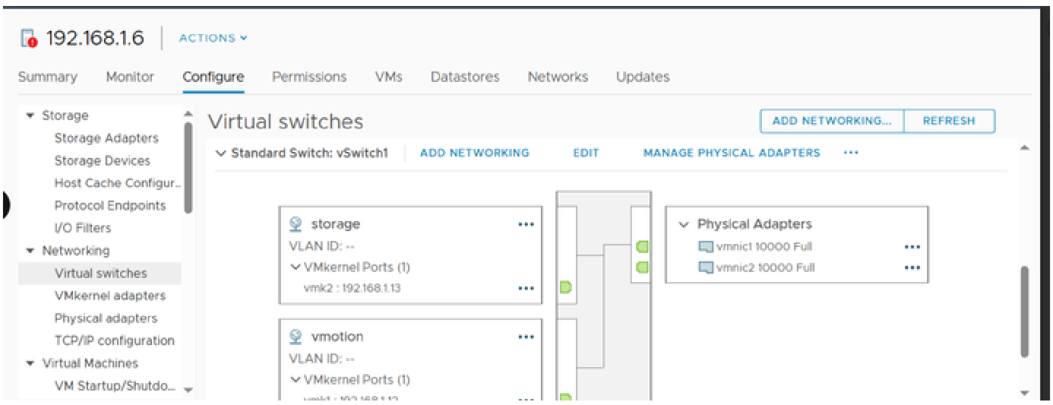
Virtual switches were configured on both ESXi hosts to separate and manage different types of network traffic. Dedicated vSwitches were assigned for migration, storage, and management to ensure better performance, security, and traffic isolation across the environment. Additionally, VMkernel port groups and VM port groups were created to enable host-level services (such as vMotion and storage access) and virtual machine connectivity.



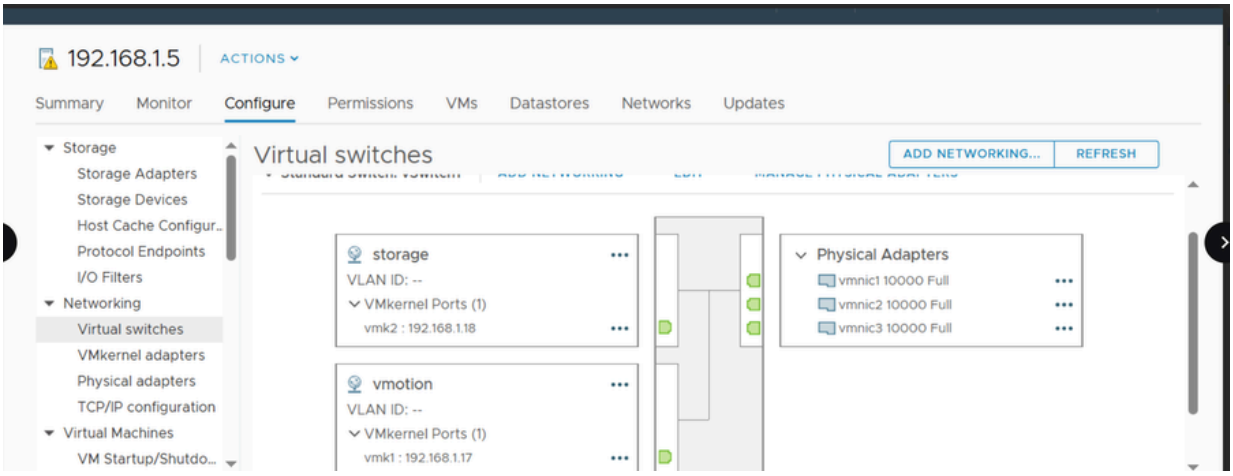
6.1 vSwitch1 for Management and vms (ESXi 1) :



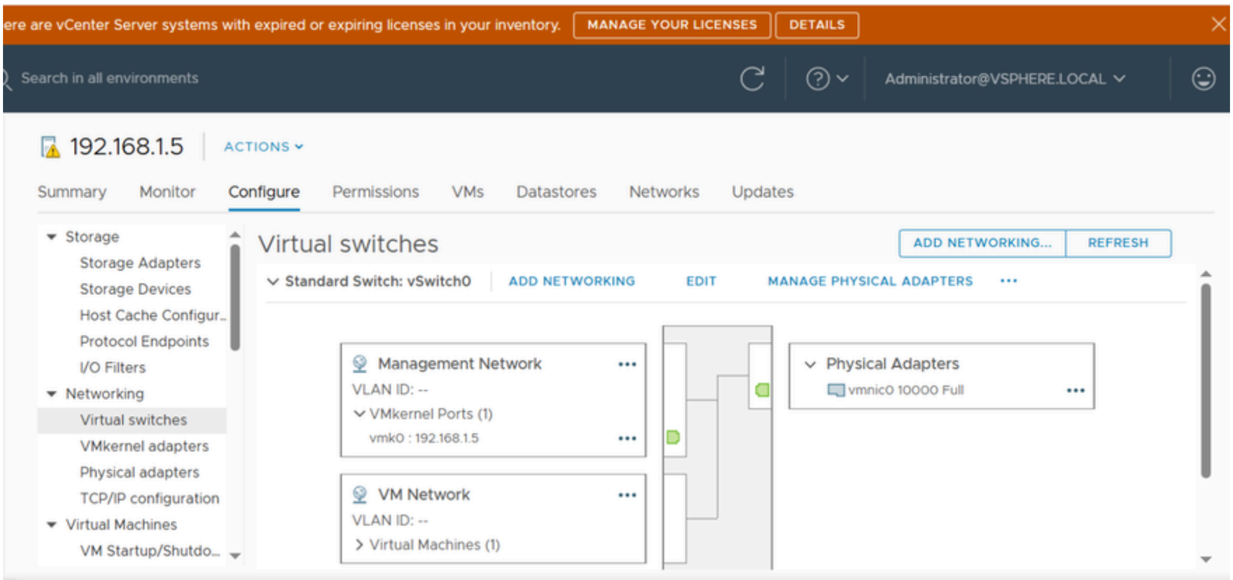
6.2 vSwitch2 for vmotion and Storage (ESXi 1) :



6.3 vSwitch2 for Migration and Storage (ESXi 2) :

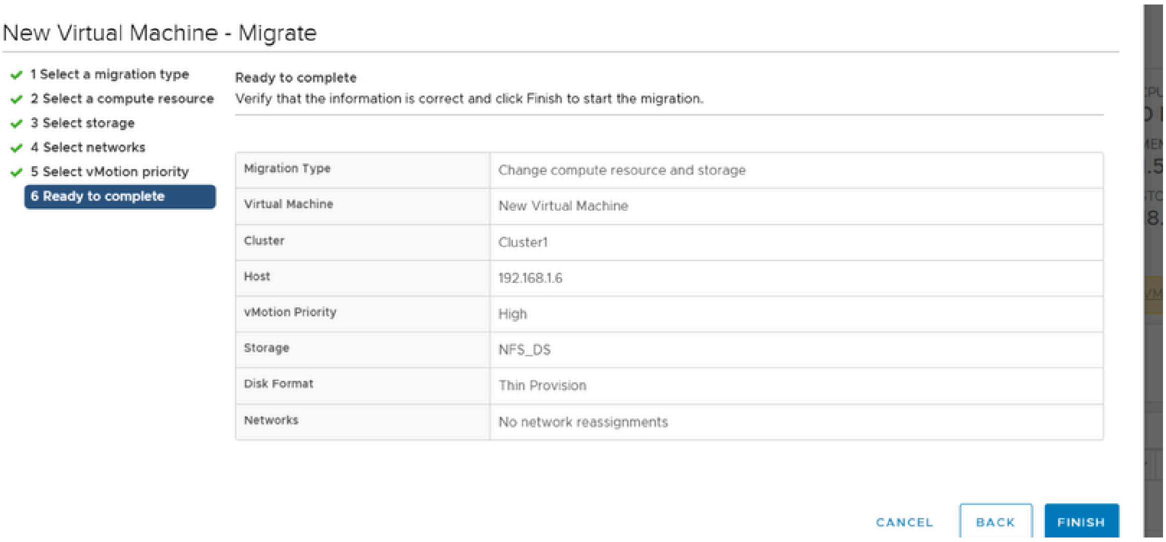


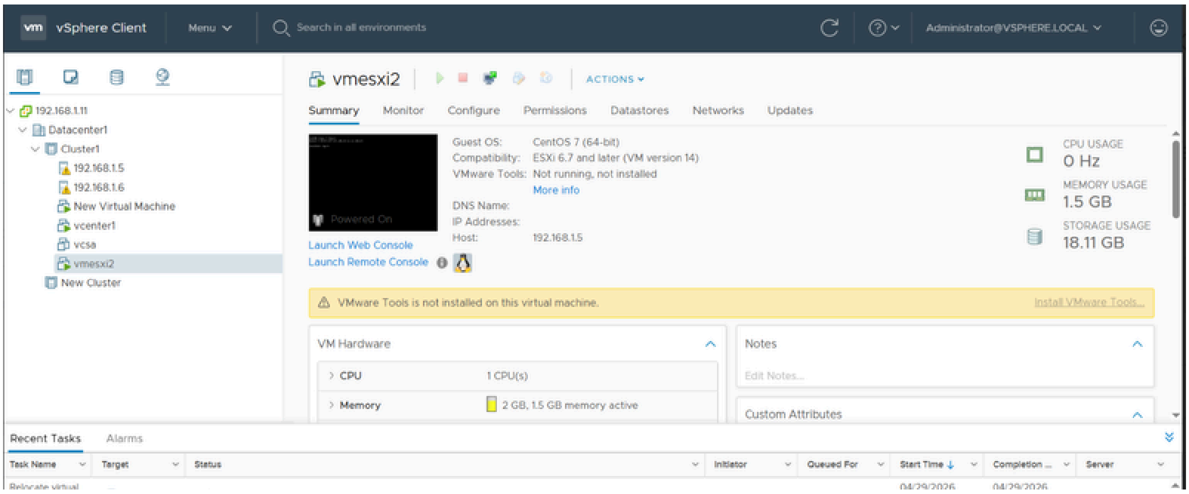
6.4 vSwitch2 for Management and vm (ESXi 2) :

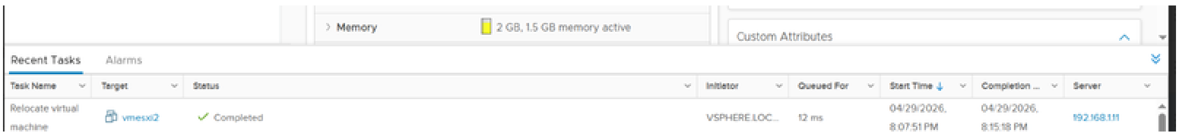


7.Migrate VMs between ESXi hosts

A virtual machine was migrated from ESXi 1 to ESXi 2 to demonstrate seamless workload movement between hosts. The VM was stored on a shared NFS datastore, which ensures that both ESXi hosts can access the same storage. This provides high availability and allows the VM to remain accessible even if one of the ESXi hosts fails.







successful migration event

8.Configuring an NFS Datastore

8.1 configure nfs server

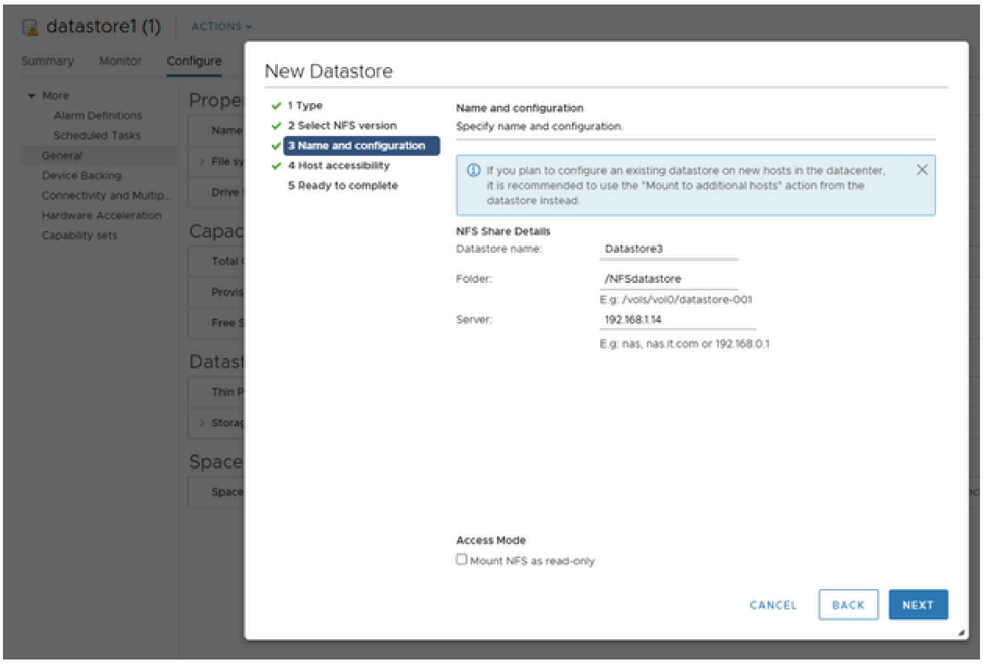
```
root@alaa:~ — nano /etc/exports

  GNU nano 5.6.1                    /etc/exports
/NFSdatastore *(rw,sync)
```

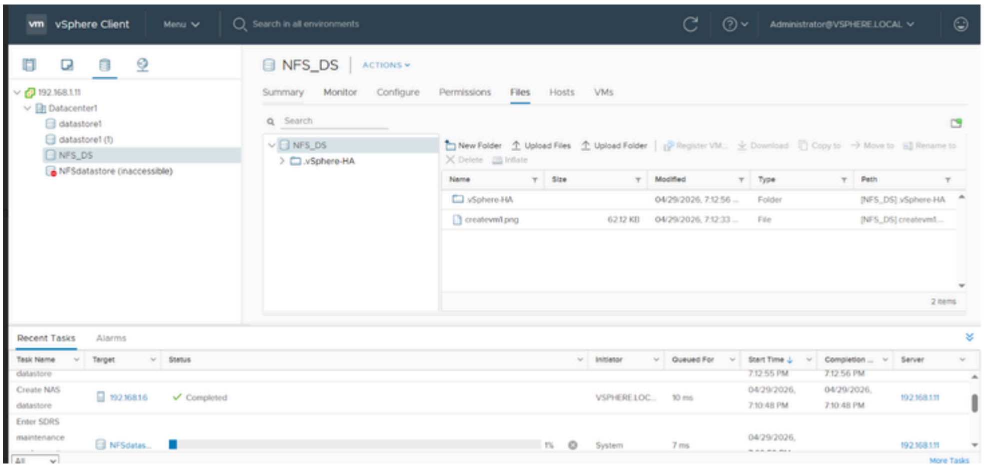
```
root@alaa:~

[root@serverb ~]# firewall-cmd --permanent --add-service=nfs
success
[root@serverb ~]# firewall-cmd --reload
success
[root@serverb ~]# firewall-cmd --list-all
public (active)
  target: default
  icmp-block-inversion: no
  interfaces: ens160
  sources:
  services: cockpit dhcpv6-client nfs ssh
```

```
[root@serverb ~]# ls /NFSdatastore/
createvm1.png  tempelate  vm2esxi2  vmesxi2
[root@serverb ~]#
```

8.2 Creating NFS data Store

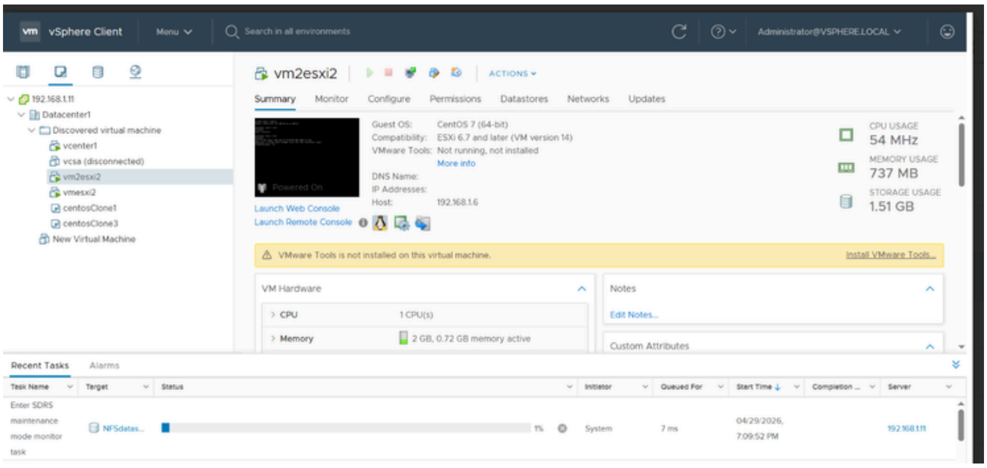
An NFS datastore was created using NFS version 3 to provide shared storage accessible by the ESXi hosts. This allows centralized and manageable storage for virtual machines, enabling features such as migration and high availability. By integrating the NFS datastore with ESXi, storage resources can be efficiently shared and managed across the virtualization environment.

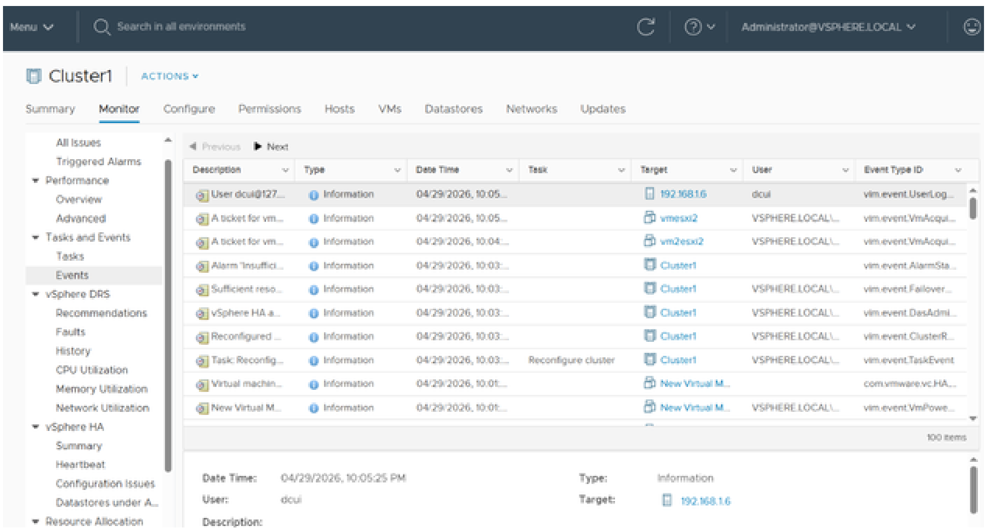




9.Testing High Availability (HA) and DRS

High Availability (HA) and Distributed Resource Scheduling (DRS) were enabled on the cluster to ensure continuous availability and efficient resource usage. A failover test was performed by shutting down one ESXi host that was running a virtual machine. As a result, HA automatically restarted and moved the VM to another available ESXi host.



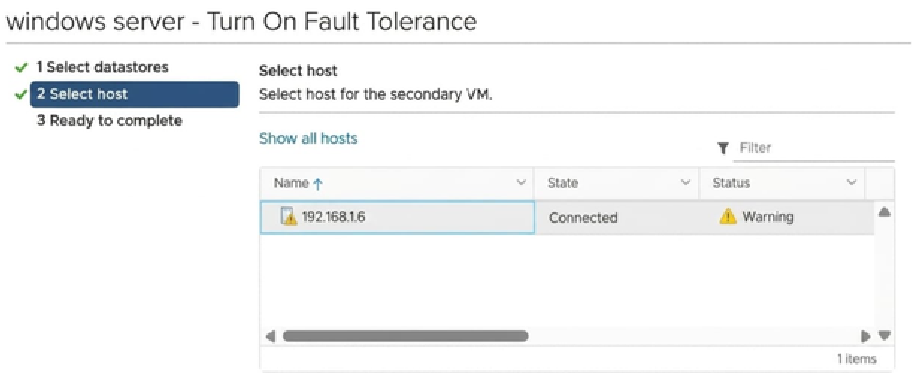


10.Configure Fault Tolerance (FT) for a Virtual Machine

A secondary (shadow) VM was created on another ESXi host to mirror the primary VM in real time. we caused esxi2 that contains the vm to fail , the secondary VM immediately on esxi1 takes over without service interruption, ensuring high reliability and continuous operation ofthe virtual machine.

10.1 Trun on Fault Tolerance

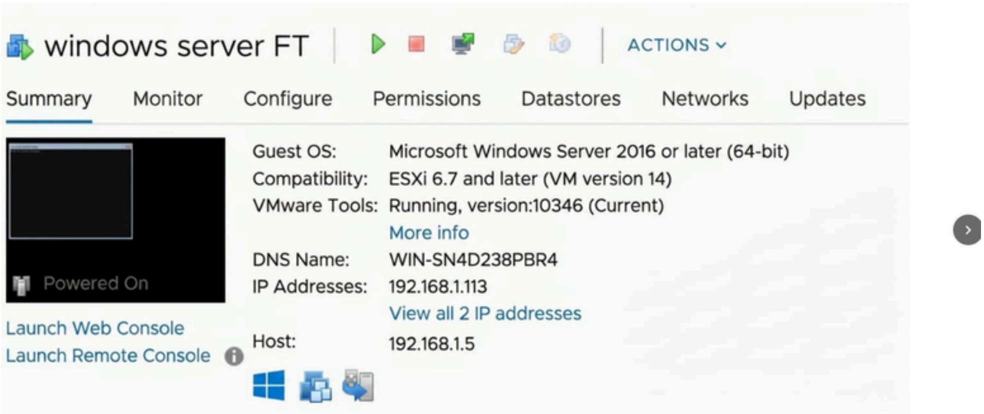
on both esxis vswitchs turn on
the FT and vmotion
on the vmkernal port



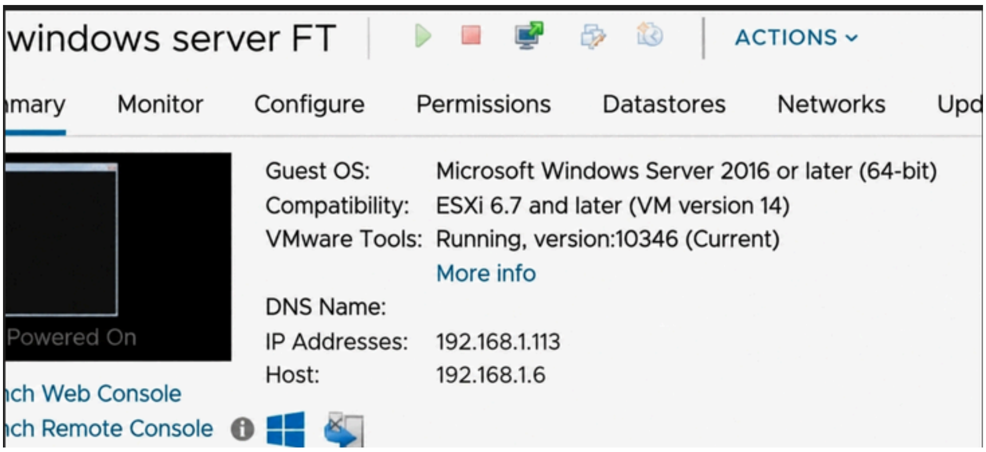
choose the secondary host for
the vm

10.2 Test FT by shutting down esxi2 that runs the windows server vm

before shutting down esxi2



after shutting down esxi2



11.Conclusion

This VMware vSphere project successfully demonstrated the deployment and configuration of a virtualized data center environment. It included setting up ESXi hosts, configuring vCenter Server, creating a cluster with HA and DRS, and implementing shared NFS storage. The project also covered virtual machine lifecycle operations, virtual networking, and VM migration between hosts. Overall, it provided hands-on experience with enterprise virtualization technologies, improving understanding of resource management, high availability, and infrastructure optimization in a virtualized environment